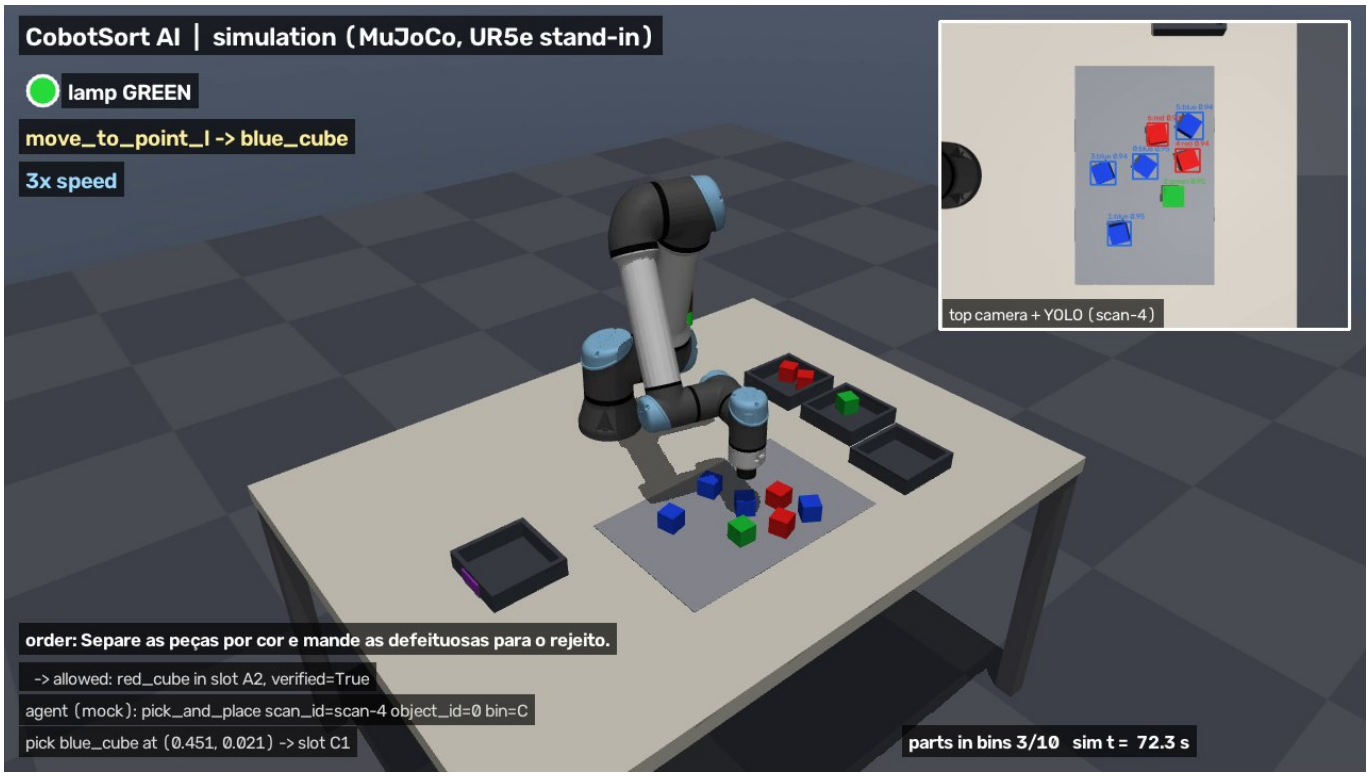
PORTFOLIO

Victor Barreto Andrade Gonçalves

Applicant for the skill **Development and Implementation of Industrial AI Solutions** — BRICS Hi-Tech Future Skills Competition 2026



CobotSort AI: the cobot sorts cubes by color after a work order in Portuguese. Top right: the top camera with YOLO detections. Bottom: the agent's validated actions.

This portfolio shows the projects behind my CV, with images and measured results. The first project, CobotSort AI, rebuilds in simulation the four modules of this skill's competition. The others show the same building blocks on real hardware and with real users: computer vision on a robot (Leak Track), an AI agent deployed with guardrails (EduBot), and deep learning at scale (HPC internship).

Contents

Project	Year	What it shows for this skill	Page
CobotSort AI	2026	All four competition modules: GUI for semi-automatic sorting (A), automatic sorting (B), training AI object detection (V), integrating it into the cell (G). Safety first.	2–3
EduBot	2026	An AI agent deployed in a full system, with guardrails, a fallback and a human approval step. Team lead; HACKaSTONE 2026 Grand Finalist; lead-author paper.	4
Leak Track	2024	Computer vision (CNN on thermal images) running on a ROS 2 robot for industrial inspection.	5
Other work	2025–26	Chest X-ray CNN, AWS DeepRacer (reinforcement learning), CatchNet (decision support for utilities), HPC internship.	5

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CobotSort AI — simulated collaborative-robot sorting cell

Personal project · October 2026 · Python, MuJoCo, YOLO11n, PyQt6 · github.com/VictorBarretoAndrade/BRICS_CobotSort

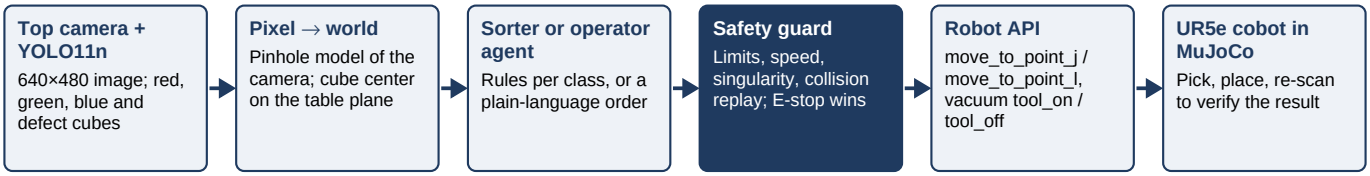
> **Why it matters for this skill:** I built it after the Skills Camp to practice every module of the competition end to end, and I report the results in the camp's judging order: safety, degree of automation, precision, speed.

A 6-axis cobot with a vacuum gripper sorts colored cubes into bins. The UR5e model stands in for the ARM95 used in the camp (same class: 6 axes, 5 kg payload). Purple cubes are defects and go to a reject bin. **Everything runs in simulation**; no number below comes from real hardware.

The four competition modules, rebuilt

Module	In CobotSort
A — GUI, semi-automatic sorting	PyQt6 operator GUI: pick a detected part and a bin, then execute. Manual jog in joint and Cartesian space, vacuum on/off, home.
B — automatic sorting	Park the arm, scan, detect, pick nearest-first, place, re-scan to verify the result; repeat until no sortable part is left.
V — train AI object detection	YOLO11n trained only on synthetic images, labeled automatically from segmentation renders, with randomized lighting, colors, camera pose and distractors.
G — integrate the AI	Detections become table coordinates through a pinhole camera model and drive the pick-and-place.

How a part moves through the cell



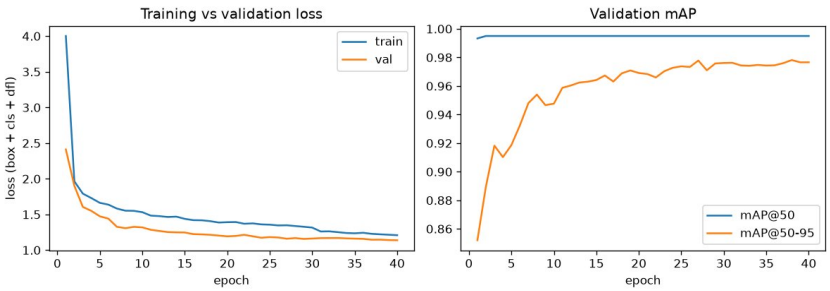
The safety guard checks the whole trajectory before the robot moves: joint limits, workspace box, collaborative speed limits (0.5 rad/s, 0.25 m/s at the tool), the manipulability index against singularities, and a collision replay. A runtime monitor stops the robot on any unplanned contact.

Kinematics, as taught in the camp

- **Forward kinematics with DH parameters**, checked against the simulator on 50 random configurations: 0.80 mm RMS position error with Universal Robots' published DH table.
- **Inverse kinematics** (damped least squares, several starting points): 100 of 100 reachable targets solved with error below 0.01 mm, on average 7.5 different joint solutions per target. An unreachable target returns no solution, unlike forward kinematics, which always has exactly one.
- **Singularities**: motions are blocked when the manipulability index drops below half the lowest value seen in normal operation (calibrated on 22,386 samples).

Detector: YOLO11n vs. a classical HSV baseline

Detector	Test set	Precision	Recall	mAP@50	Localization error mean / p95	CPU time
HSV	Synthetic test (120 images)	1.000	0.977	0.992	1.64 / 3.93 mm	2.1 ms
HSV	Hard lighting, never seen (150)	0.989	0.927	0.942	1.97 / 5.42 mm	2.1 ms
YOLO11n	Synthetic test (120 images)	1.000	0.998	1.000	1.45 / 1.65 mm	22.1 ms
YOLO11n	**Hard lighting, never seen (150)**	**0.989**	**0.978**	**0.991**	**1.38 / 2.37 mm**	**21.8 ms**



YOLO11n training curves: validation loss falls together with training loss, so there is no sign of overfitting; validation mAP@50 stays near 0.99. The hard-lighting set uses colored and stronger light that the training images never had. The cubes are small in the image (about 25–30 px), a known weakness of one-stage detectors; a two-stage detector such as Faster R-CNN is the planned comparison.

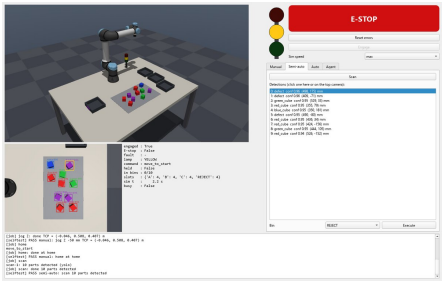
CobotSort AI — results and operator interface

Benchmark: 20 automatic episodes, 176 cubes, in the camp's judging order

Detector	Unplanned contacts	Unsafe requests blocked	Sorted without operator	Wrong bin	Placement error mean / p95	Sim. time per cube
Oracle (ground truth)	0	10/10	97.2%	0%	0.08 / 0.18 mm	24.0 s
HSV (classical)	0	10/10	97.2%	0%	1.30 / 1.90 mm	24.0 s
YOLO11n (mine)	**0**	**10/10**	**97.2%**	**0%**	**0.91 / 1.29 mm**	**24.0 s**

The 2.8% not sorted are 5 cubes whose bin was already full (4 slots per bin); the cell leaves them in place and flags them for the operator. Command: cobotsort benchmark --seeds 20.

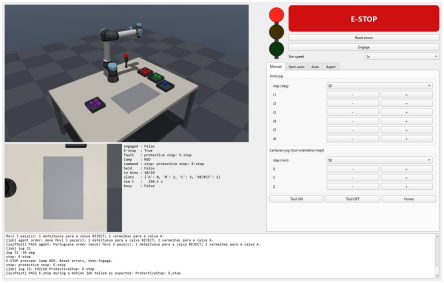
Operator GUI (module A) and safety



Semi-automatic: the operator picks a detected part and a bin.



Agent tab: a Portuguese work order, executed and summarized.



E-stop: lamp RED, then Reset errors → Engage, as in the camp's GUI.

- **Operator agent** (same pattern as EduBot: the AI decides *what*, deterministic code decides *how* and *whether it is safe*): 10 of 10 test orders in Portuguese and English passed, including refusing a bin that does not exist, asking back on ambiguous orders and stopping on "pare". Measured with the deterministic provider; the Claude provider uses the same tools and guardrails.

Skills Camp topics → where they are in the code

Camp block	Topic	In CobotSort
Collaborative robotics	Degrees of freedom, forward kinematics with DH parameters	kinematics/dh.py
	Inverse kinematics (several solutions or none)	kinematics/ik.py
	Singularities and the manipulability index	kinematics/manipulability.py, safety/guard.py
	Dynamics: $\tau = M(q)\cdot q'' + C(q, q')\cdot q' + G(q)$	cobotsort diag (gravity and Coriolis torques)
	Cobot safety: E-stop, signal lamp, reduced speed	safety/
Programming and control	MoveJ vs. MoveL and the pick-and-place sequence	motion/, robot/api.py
	Python OOP: abstract classes, dataclasses, exceptions	robot/backend.py, perception/detector.py
	Threads and Lock (GUI responsive while the robot moves)	gui/app.py
	Git flow	one branch per milestone, --no-ff merges, tags v0.0–v0.7
Neural networks	HSV thresholds, morphology, contours	perception/hsv.py
	CNN/YOLO, labeling, train/val/test, augmentation, overfitting, precision/recall/mAP	perception/

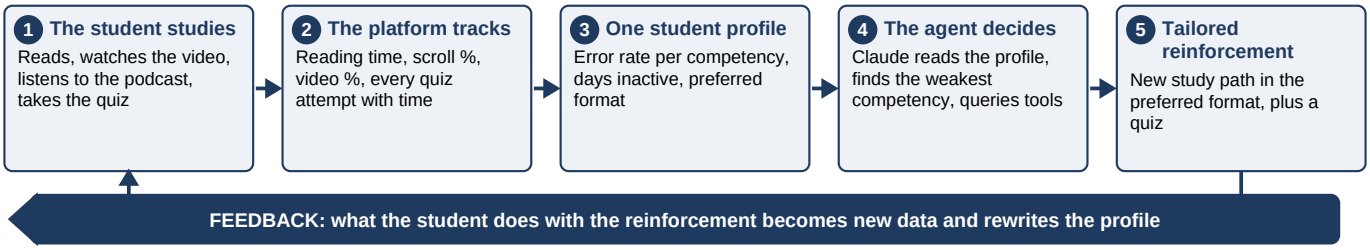
EduBot — AI agent for personalized learning

Project Lead and Tech Lead, team of 8 · 2026 · Python/Flask, MySQL, React + TypeScript, Claude via AWS Bedrock · github.com/VictorBarretoAndrade/EduBot

> **Why it matters for this skill:** It is an AI agent working inside a real system: it acts through controlled tools, its inputs are validated, a rule-based fallback takes over if the model fails, and high-impact actions wait for human approval. CobotSort's operator agent reuses this design.

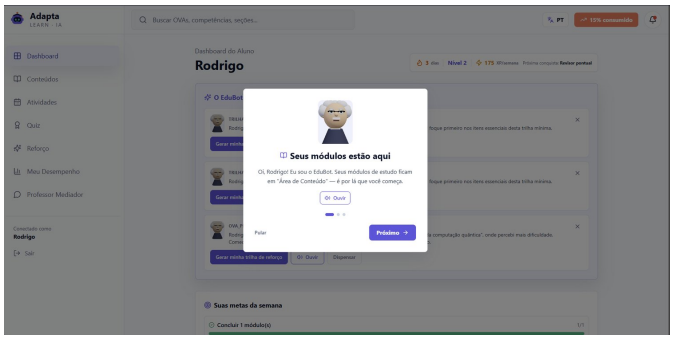
Recognition: Grand Finalist of HACKaSTONE 2026, a global contest on AI for education (Grand Final in Amsterdam). Lead-author paper on EduBot accepted at the Conference AI for Education The Netherlands 2026: "What Instantiation Reveals: Building an Agentic Remediation Loop on Top of a Conceptual Model for AI-Enhanced Virtual Learning Environments."

The data loop: student behavior feeds the AI

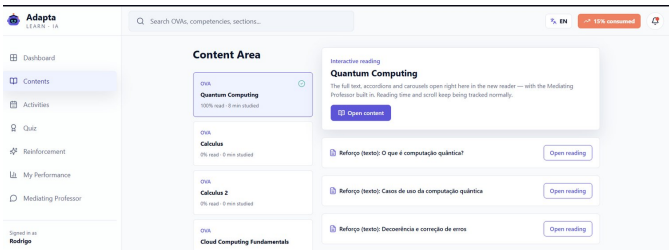


Teachers keep the decisions: a class dashboard with indicators and an alerts center, and an approval queue where high-impact agent actions wait for the teacher.

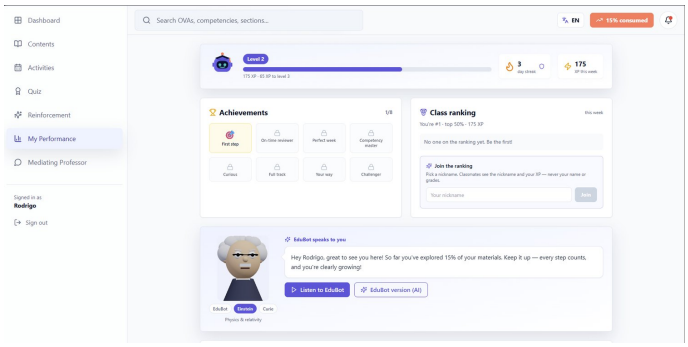
The platform (runs in Portuguese and English)



Student dashboard: EduBot welcomes the student and proposes reinforcement paths.



Content area: reading progress is tracked, and reinforcement texts appear for the weak topic.



Progress page: level, achievements, class ranking by nickname, and the EduBot avatar.

What I did as project lead and tech lead

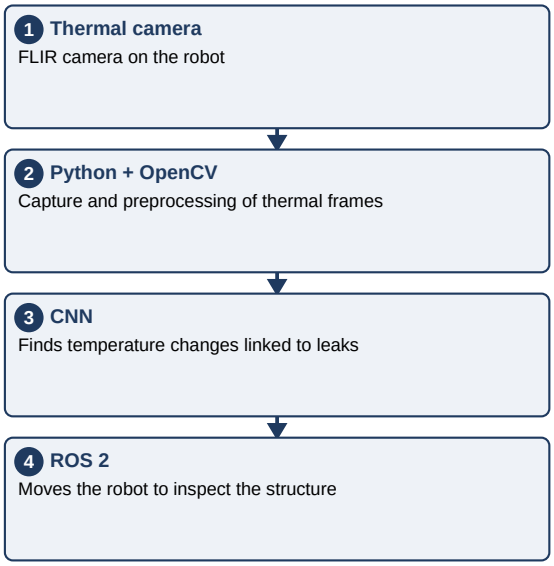
- Defined the architecture and integrated back end, front end and AI layer.
- Tool-use loop on the Anthropic API (Claude via AWS Bedrock) behind an interchangeable provider layer (mock / Bedrock / direct API).
- Guardrails: the student's identity comes only from the session token; record IDs are validated before any tool runs.
- Python/Flask REST API with 67 endpoints, MySQL with 31 tables, React + TypeScript front end, Docker Compose, 332 automated tests.

Leak Track — autonomous industrial inspection robot

IEEE Robotics and Automation Society, SENAI CIMATEC · Trainee, March – December 2024 · ROS 2, Python, OpenCV, CNNs, FLIR thermal camera

> **Why it matters for this skill:** It is industrial AI on real hardware: a neural network reads a camera on a robot and the result drives an inspection task in places that are dangerous or hard to reach for people.



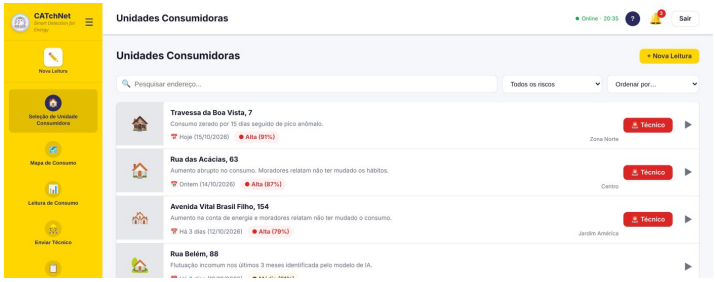


In the lab: collecting thermal images for the dataset with the thermal camera (in hand) and our Python capture script (on screen).

- Co-developed, as a trainee, a robot that detects leaks in industrial structures: ROS 2 control, Python and OpenCV image processing, and CNNs on FLIR thermal images to find temperature changes linked to leaks in real time.
- Ran the thermal-image capture experiments shown above to build the dataset.

Other work

Chest X-ray diagnosis with CNNs IEEE RAS · Project Advisor · 2025–2026	Guided a multidisciplinary team that built a CNN separating healthy lungs from pneumonia and tuberculosis cases, from data curation and validation to a results interface. Next stage: object detection (Faster R-CNN) on the RSNA dataset.
AWS DeepRacer — reinforcement learning 2025	Trained reinforcement learning models for an autonomous race car. 6th place in the Brazilian National Final (CSBC 2025) and 3rd place in the Northeast semifinal.
HPC internship SENAI CIMATEC · October 2025 – present	Deep learning (PyTorch, TensorFlow) for molecular simulation and drug discovery, trained on a SLURM-managed GPU cluster with parallel jobs and GPU allocation.



CatchNet — decision support for power distribution networks

Co-author · 2026 · poster at SIINTEC 2026 · github.com/VictorBarretoAndrade/CatchNet

Specification and prototype of an application that helps electricity utilities manage non-technical losses (energy theft and fraud). Consumer units are ranked by risk so field teams inspect the right places first; designed to plug in a machine learning model that scores theft probability.

CatchNet prototype (Portuguese interface): consumer units sorted by risk, with one-click dispatch of a field technician.